

The Biot–Savart Ring-Current Calculator

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1 What the calculator computes

The Biot–Savart calculator describes the magnetic field generated by an aromatic ring current, using the Johnson–Bovey two-loop model of the π -electron circulation. Aromatic electrons respond to an applied magnetic field by circulating around the ring. That circulation produces a secondary magnetic field at a nearby nucleus, and the secondary field changes the nuclear shielding. The observable connected to this calculation is the NMR chemical shift through its dependence on the shielding tensor.

The implementation does not solve for the electronic current from quantum mechanics. It builds a geometry-derived tensor kernel from ring vertices, ring normals, and fixed ring-type parameters. The file names use `shielding`, but the code path writes the unit-current geometric kernel. Calibration against a DFT shielding reference, or multiplication by an explicit ring-current coefficient outside this routine, is what turns the kernel into a shielding contribution in ppm.

2 Where shielding enters

For a nucleus in an applied field B_0 , the local field is written

$$B_{\text{nuc}} = (I - \sigma)B_0. \quad (1)$$

I is the 3×3 identity matrix. The shielding tensor σ is dimensionless: its ab entry says how a field applied along Cartesian direction b changes the field component felt by the nucleus along direction a . In component form,

$$B_a^{\text{sec}} = - \sum_b \sigma_{ab} B_{0,b}. \quad (2)$$

The minus sign is the NMR shielding convention: positive shielding reduces the field felt by the nucleus.

The quantum origin of σ is electronic linear response. The applied field perturbs the molecular electrons, the perturbed electrons create their own magnetic field at the nucleus, and Eq. (2) records the part proportional to B_0 . A scalar is not enough because a molecule has orientation. Turning the molecule with respect to B_0 can change which component of the electronic response is excited and which component of the secondary field arrives at the nucleus.

For an aromatic ring, this calculator keeps the part of that response that can be represented as a classical current loop. If $\hat{n}$ is the unit normal to the ring plane, then the model assumes that the ring current is driven by the normal component $\hat{n} \cdot B_0$ of the applied field. The field produced by that current is then evaluated at the target atom. This is an exact two-stage linear map within the model: first a one-dimensional meter reads the projection of B_0 onto $\hat{n}$; then that single reading drives a fixed three-component field pattern at the atom. The 3×3 tensor is the matrix of that projection followed by that field pattern.

3 The tensor split used by the code

Any real 3×3 tensor X is decomposed by `SphericalTensor::Decompose` into $T_0 + T_1 + T_2$. The scalar part is

$$T_0 = \frac{1}{3} \text{tr } X. \quad (3)$$

It is the isotropic component. In rapidly tumbling solution, this trace average is the part that enters the usual isotropic chemical shift after reference subtraction.

The antisymmetric part is stored as the three-component vector

$$\mathbf{T}_1 = \frac{1}{2} \begin{pmatrix} X_{yz} - X_{zy} \\ X_{zx} - X_{xz} \\ X_{xy} - X_{yx} \end{pmatrix}. \quad (4)$$

This vector is the Levi-Civita dual of the antisymmetric matrix. It is usually not observed directly in standard isotropic NMR, but it is present whenever the response matrix is not symmetric.

The remaining part is the symmetric traceless matrix

$$\mathbf{S} = \frac{\mathbf{X} + \mathbf{X}^\top}{2} - T_0 \mathbf{I}, \quad \text{tr } \mathbf{S} = 0. \quad (5)$$

It has five independent components. The code stores those five entries in an isometric real spherical basis,

$$\mathbf{T}_2 = \begin{pmatrix} \sqrt{2} S_{xy} \\ \sqrt{2} S_{yz} \\ \sqrt{3/2} S_{zz} \\ \sqrt{2} S_{xz} \\ (S_{xx} - S_{yy})/\sqrt{2} \end{pmatrix}. \quad (6)$$

“Isometric” means that the five-vector length equals the Frobenius length of the underlying symmetric traceless matrix: $\sum_m T_{2m}^2 = \sum_{ab} S_{ab}^2$. The calculator writes all three parts in the packed order

$$[T_0, T_{1x}, T_{1y}, T_{1z}, T_{2,-2}, T_{2,-1}, T_{2,0}, T_{2,+1}, T_{2,+2}].$$

The Biot–Savart kernel is rank one, but it is generally not symmetric, so T_0 , T_1 , and T_2 can all be nonzero.

4 The method implemented

Each aromatic ring supplies a vertex list $\mathbf{v}_i$ in Å, a center, an average radius, and a unit normal $\hat{\mathbf{n}}$. The center is the mean of the vertices. The normal is the right singular vector of the centered vertex matrix with the smallest singular value, with its sign chosen to match the cyclic vertex order. This is the least-squares plane normal: it points perpendicular to the least-squares plane through the ring atoms.

The finite-wire field used for each polygon edge is

$$\mathbf{F}(\mathbf{a}, \mathbf{b}, I, \mathbf{r}) = \frac{\mu_0}{4\pi} \frac{I}{|\boldsymbol{\ell} \times \mathbf{d}_A|^2} \left(\frac{\boldsymbol{\ell} \cdot \mathbf{d}_A}{|\mathbf{d}_A|} - \frac{\boldsymbol{\ell} \cdot \mathbf{d}_B}{|\mathbf{d}_B|} \right) (\boldsymbol{\ell} \times \mathbf{d}_A). \quad (7)$$

Here $\mathbf{a}$ and $\mathbf{b}$ are the segment endpoints in metres, $\mathbf{r}$ is the field point in metres, $\boldsymbol{\ell} = \mathbf{b} - \mathbf{a}$, $\mathbf{d}_A = \mathbf{r} - \mathbf{a}$, and $\mathbf{d}_B = \mathbf{r} - \mathbf{b}$. The current I is in amperes and $\mathbf{F}$ is in tesla. The prefactor used by the code is $\mu_0/(4\pi) = 1 \times 10^{-7} \text{ T m/A}$, the pre-2019 SI value. If the field point is within $1 \times 10^{-25} \text{ m}$ of an endpoint, or if $|\boldsymbol{\ell} \times \mathbf{d}_A|^2 < 1 \times 10^{-70} \text{ m}^2$, the segment returns zero rather than evaluating the singular expression.

The Johnson–Bovey model replaces the π density by two identical polygonal loops, one at $+d_t \hat{\mathbf{n}}$ and one at $-d_t \hat{\mathbf{n}}$ from the ring plane. Writing the Å-to-metre conversion implicitly, the unit-current field at atom position $\mathbf{r}$ for ring type t is

$$\mathbf{B}_t^{(1)}(\mathbf{r}) = \sum_i \left[\mathbf{F}(\mathbf{v}_i + d_t \hat{\mathbf{n}}, \mathbf{v}_{i+1} + d_t \hat{\mathbf{n}}, \frac{1}{2} \text{ nA}, \mathbf{r}) + \mathbf{F}(\mathbf{v}_i - d_t \hat{\mathbf{n}}, \mathbf{v}_{i+1} - d_t \hat{\mathbf{n}}, \frac{1}{2} \text{ nA}, \mathbf{r}) \right]. \quad (8)$$

The code converts Å to metres and nanoamperes to amperes at the boundary of this calculation. The polygon is not a numerical quadrature mesh; its edges are the physical ring-wire approximation, and each edge is integrated by the closed form in Eq. (7).

The tensor kernel stored for that ring and atom is

$$G_{ab}^{(t)}(\mathbf{r}) = -10^6 \hat{n}_b B_{t,a}^{(1)}(\mathbf{r}). \quad (9)$$

The row index a is the component of the secondary field, and the column index b is the component of the applied field. Equation (9) is the outer product $-B_t^{(1)} \hat{\mathbf{n}}^T$, scaled by 10^6 . Its trace gives

$$T_0^{(t)} = -\frac{10^6}{3} \hat{\mathbf{n}} \cdot \mathbf{B}_t^{(1)}. \quad (10)$$

If a ring-current coefficient κ_t in nA/T is applied later, the usual shielding contribution would be $\kappa_t G^{(t)}$ in ppm. The present `Compute()` path records κ_t for provenance but sums Eq. (9) with a unit current for every accepted ring.

Ring type	d_t (Å)	κ_t (nA/T)
PHE benzene	0.64	-12.00
TYR phenol	0.64	-11.28
TRP benzene, TRP6	0.64	-12.48
TRP pyrrole, TRP5	0.52	-6.72
TRP indole perimeter, TRP9	0.60	-19.20
HIS, HID, HIE imidazole	0.50	-5.16

For each atom, the calculator queries aromatic rings within 15.0 Å. A candidate is rejected when its center distance is less than 0.1 Å, when that distance is not greater than the source radius $0.5(2R) = R$, or when the atom is a ring atom or covalently bonded to a ring atom. These filters mark where the through-space loop model is not being used: at singular distances, inside the current distribution, and in the through-bond region. Accepted ring tensors are summed,

$$G_i = \sum_{r \in \mathcal{A}_i} G^{(t_r)}(\mathbf{r}_i), \quad (11)$$

where $\mathcal{A}_i$ is the set of accepted nearby aromatic rings for atom i , and t_r is ring r 's type. The sum is decomposed by Eqs. (3)–(6). The code also accumulates per-ring-type T_0 and T_2 , stores the total unit-current $\mathbf{B}$ field, and records accepted-ring counts within 3 Å, 5 Å, 8 Å, and 12 Å shells.

There is also a free-point sampling path. It evaluates the same unit-current field and kernel on a grid point, but it does not use atom-topology filters because a grid point is not a protein atom. It keeps the 0.1 Å singularity guard, skips points whose center distance is below the average ring radius, and applies the 15.0 Å cutoff.

5 Inputs and output

The calculator consumes atom coordinates in Å, typed aromatic ring topology, ring geometry, and the spatial index used to find nearby rings. It does not consume MOPAC charges, AIMNet2 charges or embeddings, ff14SB charges, APBS electrostatic fields, or DSSP secondary-structure labels; those data feed other calculators. Their internal methods are outside the scope of this document.

The written arrays are `bs_shielding.npy`, an $N \times 9$ packed $T_0 + T_1 + T_2$ kernel; `bs_per_type_T0.npy`, an $N \times 8$ per-aromatic-ring-type scalar table; `bs_per_type_T2.npy`, an $N \times 40$ table containing eight five-component T_2 blocks; `bs_total_B.npy`, an $N \times 3$ unit-current magnetic field; and `bs_ring_counts.npy`, an $N \times 4$ accepted-ring shell-count table. These arrays are geometric descriptors. Agreement with DFT is assessed by calibration and correlation, not by expecting the raw kernel to reproduce DFT shielding pointwise.